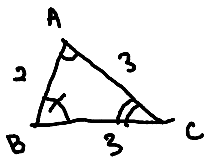
geometry

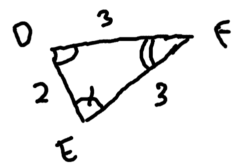
Thursday, 5 June, 2025 8:30 AM

1: CONGRUENCY + SIMILARITY OF △

congruency

↳ corresponding sides and corresponding angles are equal





↪ rotation/reflection

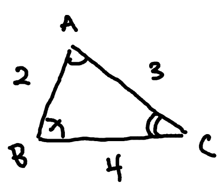
SSA ✗

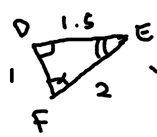
△ABC ≅ △DEF

3 criterions
- SSS → all 3 sides equal
- SAS → 2 sides + angle between are equal
- ASA → 2 angles + side are equal

Similarity

↳ corresponding angles are equal, corresponding sides are proportional



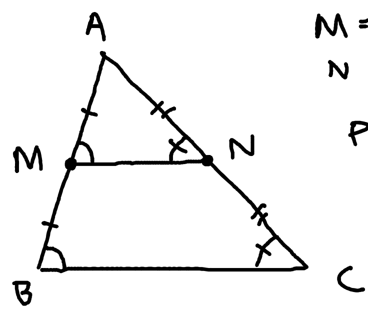


↘ rotation/reflection + scaled

$$\frac{AB}{DF} = \frac{AC}{DE} = \frac{BC}{EF}$$

△ABC ∼ △DEF

3 criterions
- AA → 2 angles equal
- SSS → sides are proportional
- SAS → 2 sides proportional + angle between equal



M = midpt AB

N = midpt AC

Prove $MN \parallel BC$ → parallel

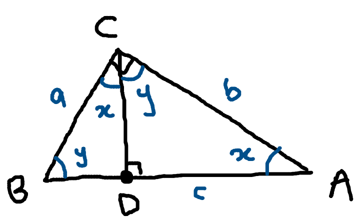
and $BC = 2MN$

by SAS, $\triangle AMN \sim \triangle ABC$

$\angle AMN = \angle ABC \Rightarrow MN \parallel BC$

$$\frac{1}{2} = \frac{AM}{AB} = \frac{MN}{BC} \Rightarrow BC = 2MN$$

Prove the Pythagorean Theorem



$a^2 + b^2 = c^2$

hint: $\frac{x}{y} = \frac{z}{x}$

$\Rightarrow x^2 = yz$

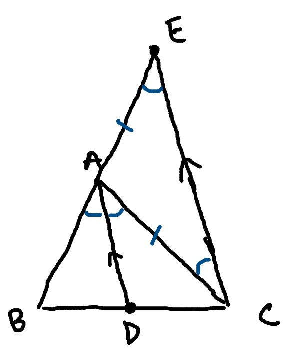
by AA, $\triangle ACD \sim \triangle CBD \sim \triangle ABC$

$$\frac{BC}{BD} = \frac{AB}{BC} \Rightarrow BC^2 = AB \times BD$$

$$\frac{CA}{AD} = \frac{AB}{CA} \Rightarrow CA^2 = AB \times AD$$

$$a^2 + b^2 = BC^2 + CA^2 = AB(BD) + AB(AD)$$
$$= AB\,(\underbrace{BD + AD}_{=AB}) = AB^2 = c^2$$

Angle Bisector Theorem



$$\frac{AB}{BD} = \frac{AC}{CD}$$

$$\frac{BE}{AB} = \frac{BC}{BD}$$

$$\Rightarrow \frac{AB + AE}{AB} = \frac{BD + CD}{BD}$$

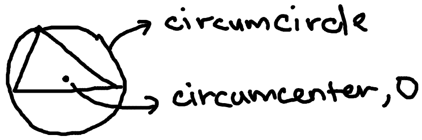
$$\Rightarrow \cancel{1} + \frac{AE}{AB} = \cancel{1} + \frac{CD}{BD}$$

$$\Rightarrow \frac{AB}{BD} = \frac{AE}{CD} = \frac{AC}{CD}$$

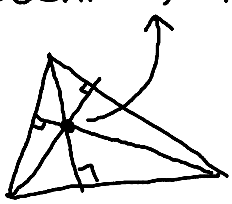
2: △ CENTERS

1) circumcenter, O

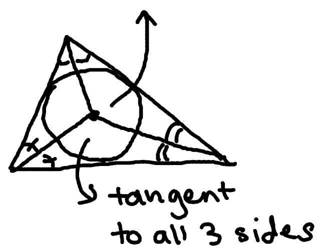
line → 2pts, circle → 3pts



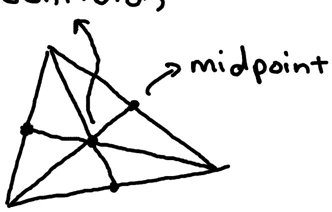
2) orthocenter, H

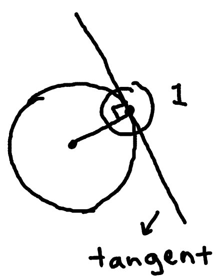


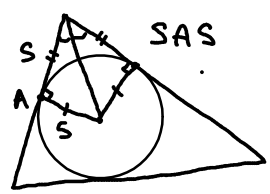
3) incenter, I



4) centroid, G

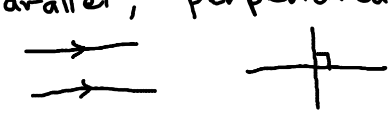






3: ANGLE CHASING

parallel, perpendicular



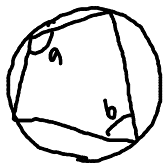
☆ angles in circles

cyclic quadrilateral

(i)

↳ 4 vertices lie on a circle

(1)



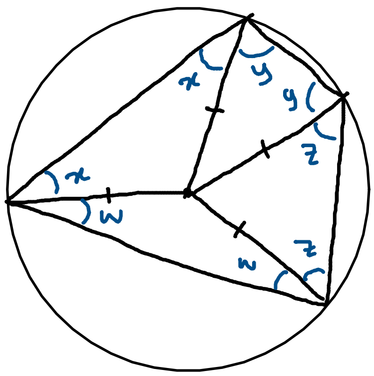
→ 4 vertices lie on a circle

$a + b = 180°$

↳ prove cyclic

isosceles

→ 2 sides of same length

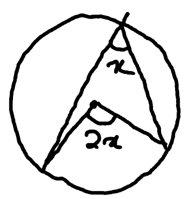


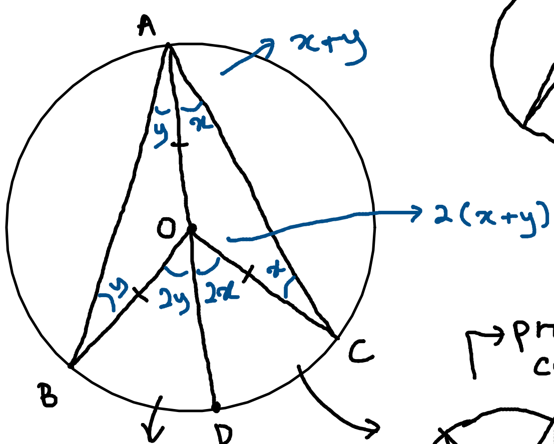
$x + y + w + z = 180°$

bcs

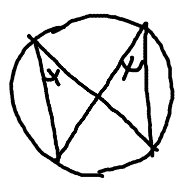
$2(x + y + w + z) = 360°$

(2)

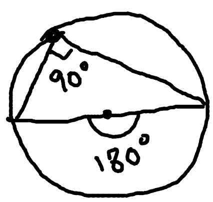




→ prove cyclic

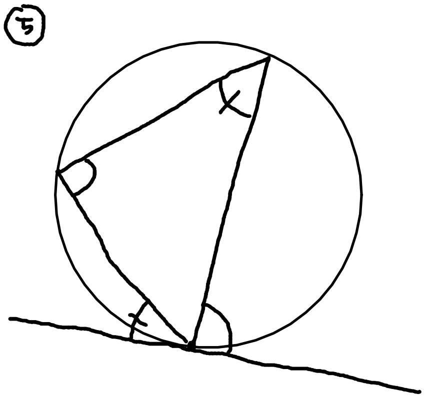


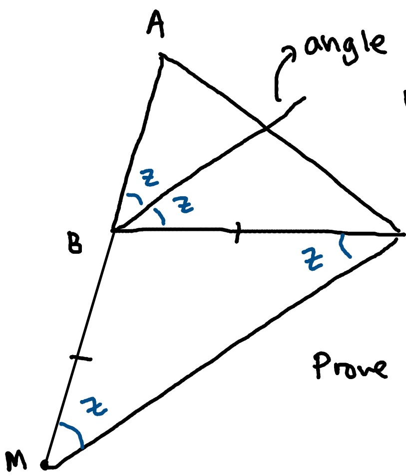
(3)



(4)

(5)



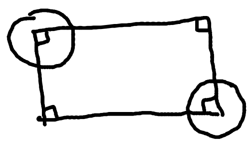


angle bis., ℓ

$BM = BC$

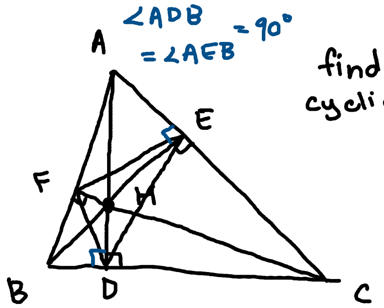
Prove $MC \parallel \ell$.

Prove all rectangles are cyclic.



sum $= 180^\circ$

$\angle ADB = \angle AEB = 90^\circ$



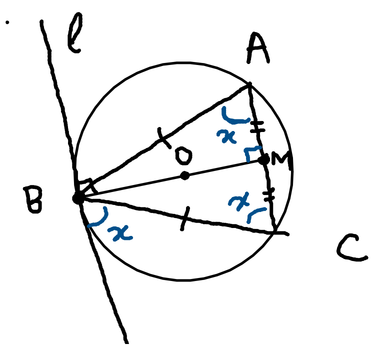
find 6 cyclic quads

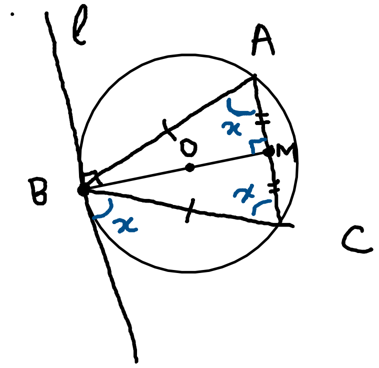
AFHE AEDB
BDHF BFEC
EHDC AFDC

$\triangle ABC$, $AB = BC$.

ℓ = tangent to (ABC) at B

Prove $\ell \parallel AC$

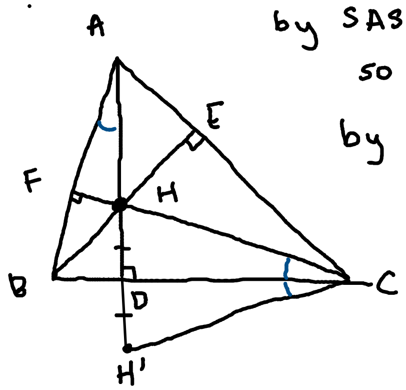




H = orthocenter of $\triangle ABC$

H' = reflection of H across BC

Prove H' lies on (ABC)



by SAS, $\triangle CDH \cong \triangle CDH'$

so $\angle BCH' = \angle BCH$

by AA, $\triangle CDH \sim \triangle AFH$,

so $\angle BCH = \angle BAH'$

$\therefore \angle BCH' = \angle BAH'$

$\Rightarrow ABH'C$ cyclic

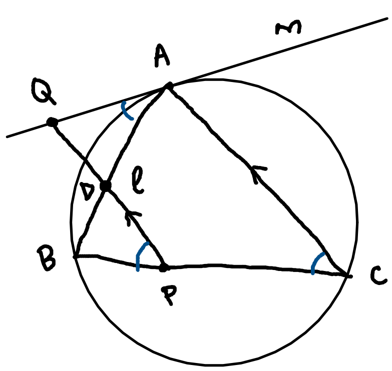
P = point on BC of $\triangle ABC$

ℓ = line parallel to AC through P

m = line tangent to (ABC) through A

$Q = \ell \cap m$

Prove $APBQ$ is cyclic



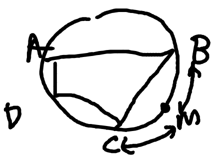
$QP \parallel AC \Rightarrow \angle QPB = \angle ACB$

tangency $\Rightarrow \angle ACB = \angle QAB$

$\therefore \angle QPB = \angle QAB$

$\Rightarrow APBQ$ cyclic

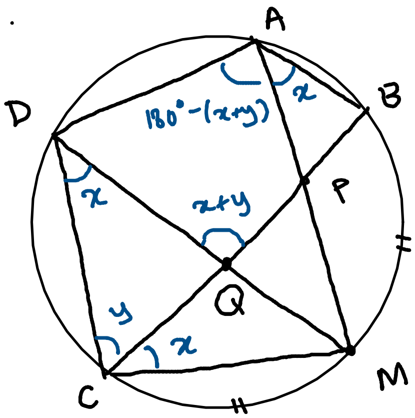
cyclic quad ABCD



M = midpt of arc BC
not containing A and D

$P = AM \cap BC$, $Q = DM \cap BC$

Prove APQD cyclic



hint: there are equal angles
↳ M = midpt $\overset{\frown}{BC}$

hint: Property 1